

policy continues to drive the electronic equipment manufacturing industry in India. Numerous global electronic equipment manufacturing companies have set up production facilities in India to take advantage of the growing domestic market, and to cater to neighbouring markets in the region. As a result, semiconductor consumption is also growing at a rapid pace," says Ganesh Ramamoorthy, research director, Gartner.

"Given the low penetration and the growing demand for key electronic equipment such as mobile phones, desktops, laptop computers and LCD TVs, we believe the Indian market will be able to easily sustain high growth rates in the coming years. Therefore we expect India's semiconductor consumption to grow the fastest across the globe through 2015 at a compound annual growth rate of 15.9 per cent to reach nearly \$15 billion. Through 2015, nearly three-fourth of India's semiconductor consumption will be accounted for by these three electronic equipment segments," Ramamoorthy added.

Nearly 2000 chips are being designed in India every year, with more than 20,000 engineers getting engaged in various aspects of chip design and verification. "With India becoming the hub for semiconductor design, the embedded industry offers tremendous job opportunities countrywide. If you plan to enter into this field, you need not worry about the respective job prospects. We require tons of embedded professionals to create all that is possible—be it in telecom, networking, automotive, medical and so on," quips

Where do I fit?

1. Software engineers handle the increasing focus on efficient and intuitive firmware designs.
2. Electrical engineers with talents in the analogue domain. As always, these digital devices need to be interfaced with an analogue world and talents in this domain will continue to be invaluable.
3. Electrical/electronics engineers excel in the digital domain. The industry not only needs talented people who can work at a high level with embedded devices, but it will always have a need for those who design these products at the gate level as well.
4. Systems engineers focus on integrating embedded devices into such applications as control systems.
5. Field applications engineers work directly with customers to help integrate a manufacturer's products into their designs.
6. Field sales engineers serve as a customer's first contact with a company's products and technologies. These engineers are crucial in developing new relationships, early on.
7. Process engineers focus on control and optimisation of areas such as fabrication of silicon devices.
8. Product marketing engineers are a unique breed in any company. They provide direction to the field while evaluating market trends to recommend new products.

Venkatesh Kumaran, country manager, Wind River India.

He also underlines the fact that the embedded sector suffers from lack of quality manpower. "There has always been a dearth of embedded professionals worldwide and in our country. One of the main reasons behind this is that our universities and engineering colleges do not put a great emphasis specifically on embedded courses/practicals. Not many experienced professionals are available in the teaching community here who can take that as a passion and help students at large in the country," he adds.

What role to choose?

Any embedded system is teamwork of three different sets of people including hardware engineers, software engi-

neers and application domain experts. You can look for a job under any one of these broad categories. If you are comfortable in the hardware domain, you will be assigned responsibility for hardware platform and module prototyping, debugging and testing. Also, you have to ensure compliance with standards and product specifications and initiate design changes as necessary.

As a software engineer, you can start as an embedded systems programmer with responsibility of tasks like analysis and optimisation of embedded software for the targeted real-time operating system (RTOS). Your activities may include developing installable and built-in device drivers, kernel modifications and embedded application.

You can begin as a wireless protocol engineer if you seek to explore the application domain. You will be responsible for the development, integration and testing of various protocols within an embedded firmware stack used in mobile devices.

Who's hiring?

Some of the major recruiters in this sector are Texas Instruments (TI), STMicroelectronics, Intel, Freescale

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